# EXPERIMENT 20

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

## K-Means Clustering

**Aim:** Group mall customers into clusters using K-Means.

## Python Program

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
df=pd.read_csv('Mall_Customers.csv')
x=df[['Annual Income (k$)','Spending Score (1-100)']]
model=KMeans(n_clusters=5,random_state=1,n_init=10)
df['Cluster']=model.fit_predict(x)
print(df.head())
plt.scatter(x.iloc[:,0],x.iloc[:,1],c=df['Cluster'])
plt.xlabel('Annual Income')
plt.ylabel('Spending Score')
plt.show()
```

## Result

Customers are assigned to five clusters and shown in a scatter plot.

**Note:** For CSV programs, keep the named CSV file in the same folder and check that column names match.